

Keerthana.p

192521289

Cloud computing – CSA1511

Short Answer Test

1. Define Cloud Computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition

Cloud Computing is the delivery of computing services such as storage, servers, software, and databases over the Internet instead of using a personal computer or local server.

Four Characteristics

1. On-Demand Self-Service

Users can access cloud services whenever they need without contacting the service provider.

Example: Creating a Google Drive account and using storage immediately.

2. Broad Network Access

Cloud services can be accessed from anywhere using the Internet through a mobile phone, laptop, or tablet.

Example: Accessing Gmail from any device.

3. Resource Pooling

The cloud provider shares computing resources among many users while keeping their data separate.

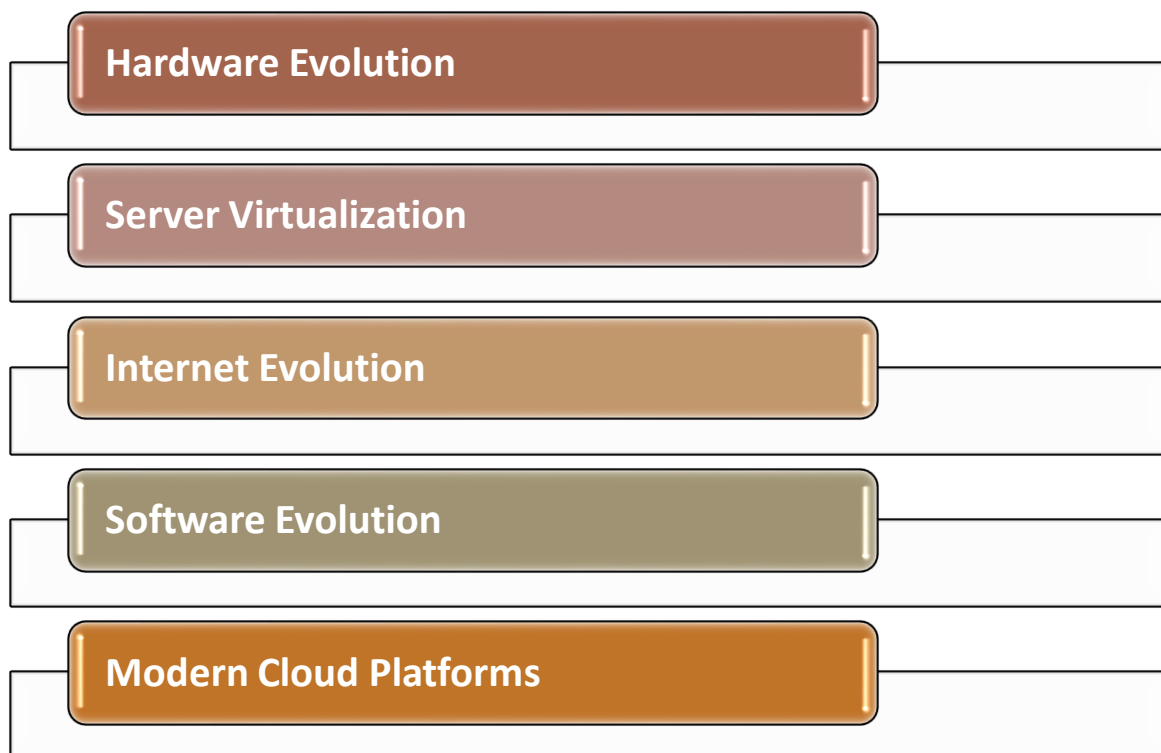
Example: Many users use AWS servers at the same time.

4. Rapid Elasticity

Cloud resources can increase or decrease based on user demand.

Example: During an online shopping sale, Amazon increases server capacity automatically.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.



Hardware Evolution

Initially, every company purchased its own computers and servers to store data **Hardware Evolution**



Server Virtualization

One physical server was divided into multiple virtual machines, improving resource utilization and reducing hardware costs.



Internet Evolution

The Internet became faster and more reliable, allowing users to access applications and data online.



Software Evolution

Applications moved from local computers to web-based software such as Gmail, Google Docs, and Microsoft 365.



Modern Cloud Platforms

Cloud providers like **AWS, Microsoft Azure, and Google Cloud** now offer storage, computing power, databases, and AI services through the Internet.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Role of Server Virtualization

Server virtualization divides one physical server into multiple virtual servers.

Each virtual server works independently and can run different operating systems and applications.

This improves:

- Resource utilization
- Cost efficiency
- Flexibility

Difference

Virtualization	Cloud Computing
An enabling technology	A service delivery model
Creates multiple virtual machines on one physical server	Provides computing services over the Internet
Focuses on efficient hardware usage	Focuses on delivering services to users
Example: VMware	Example: AWS, Azure

4. Discuss the role of data centers in delivering cloud services.

Answer

A data center is a building that contains thousands of servers, storage devices, networking equipment, and cooling systems.

Its role is to:

- Store customer data safely.
- Process user requests.
- Host cloud applications.
- Provide backup and disaster recovery.
- Ensure cloud services are available 24/7.

Example

When you save a file in Google Drive, it is stored in Google's data center.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Service Model	Meaning	Example
IaaS (Infrastructure as a Service)	Provides virtual servers, storage, and networking over the Internet.	Amazon EC2 (AWS)
PaaS (Platform as a Service)	Provides a platform for developing and deploying applications without managing hardware.	Google App Engine
SaaS (Software as a Service)	Provides ready-to-use software over the Internet.	Google Docs
XaaS (Anything as a Service)	Provides various IT services such as storage, databases, AI, and security through the cloud.	Microsoft Azure AI Services